

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

FOR

B.TECH.

in

Artificial Intelligence and Machine Learning

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]

Branch/Course: Artificial Intelligence and Machine Learning

Semester-V (Third Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AIML301	Reinforcement Learning	3+0
2.	Professional Core Course	AIML302	Database Management System	3+0
3.	Professional Core Course	AIML303	Design and Analysis of Algorithms	4+0
4.	Professional Core Course	AIML304	Data Mining and Data Warehousing	3+0
5.	Professional Core Course	AIML305	Natural Language Processing	3+0
6.	Professional Elective Courses		Elective-I (from SWAYAM/ Department of AIML)	3+0
7.	Humanities & Social Science including Management Courses	HSM301	Organization Behavior	3+0
8.	Professional Core Course	AIML352	Database Management System Lab	0+1
9.	Professional Core Course	AIML353	Design and Analysis of Algorithm Lab	0+1
10.	Project work, Seminar and internship in industry or elsewhere	AIMLMP1	AIML Mini Project – I	0+1
Total Credit				25

Reinforcement Learning (AIML301)

Credit: 3+0

Course Objective:

To understand the fundamental concepts and principles of reinforcement learning. To explore model free and model-based RL techniques, including Monte Carlo and temporal difference methods. To analyse deep reinforcement learning algorithms and their applications.

Syllabus

1. Fundamentals of Reinforcement Learning: Introduction to Reinforcement Learning (RL): Basic concepts, differences between supervised, unsupervised, and reinforcement learning, agent-environment interaction, Markov Decision Processes (MDPs), reward functions, policy vs. value function, exploration vs. exploitation. Dynamic Programming for RL: Bellman equations, policy evaluation, policy iteration, value iteration, discounted and undiscounted rewards, optimality equations.

2 Model-Free RL and Approximate Solutions: Monte Carlo and Temporal Difference Learning: Monte Carlo methods, first-visit and every-visit MC, TD(0) learning, n-step TD, SARSA, Q-learning, Expected SARSA, trade-offs between MC and TD methods. Function Approximation in RL: Need for function approximation, linear function approximation, deep Q-networks (DQN), policy gradient methods, actor-critic methods, in deep RL.

3. Advanced Reinforcement Learning Algorithms: Policy Optimization Methods: REINFORCE algorithm, trust region policy optimization (TRPO), proximal policy optimization (PPO), soft actor-critic (SAC), deep deterministic policy gradient (DDPG). Multi-Agent and Hierarchical RL: Cooperative and competitive multi-agent RL, self-play and adversarial training, hierarchical RL, options framework and game AI.

4. Applications and Future Trends in RL: Real-World Applications of RL: RL in robotics, self-driving cars, recommendation systems, healthcare, trading and finance, energy optimization, automated control systems. Ethical Considerations and Challenges: Challenges in reward shaping, sample efficiency, safety constraints in RL, ethical concerns in AI decision-making, generalization in RL models, ongoing research trends and future directions in reinforcement learning.

Course Outcome (CO): Upon completion of the course, the students should be able to

CO1: Understand the mathematical foundation and principles of reinforcement learning.

CO2: Implement fundamental RL algorithms like Q-learning, SARSA, and temporal difference learning.

CO3: Apply deep reinforcement learning methods for decision-making and control problems.

CO4: Design and optimize policy-based RL algorithms for real-world applications.

References

1. Richard S. Sutton, Andrew G. Barto – *Reinforcement Learning: An Introduction*, MIT Press, 2018.
2. David Silver's RL Course Notes – Available online, University College London (UCL).
3. Francois-Lavet, Dumoulin-Thurston, Holland, et al. – *An Introduction to Deep Reinforcement Learning*, Foundations and Trends in Machine Learning, 2018.
4. Csaba Szepesvári – *Algorithms for Reinforcement Learning*, Morgan & Claypool Publishers, 2010.
5. Nicolas Heess, Timothy P. Lillicrap, Tom Erez, Yuval Tassa, David Silver, Martin Riedmiller – *Deep Reinforcement Learning with Continuous Control*, 2015.

Database Management System (AIML302)

Credit: 3+0

Course Objectives:

To explain basic database concepts, applications, data models, schemas and instances. To demonstrate the use of constraints and relational algebra operations. To emphasize the importance of normalization in databases.

Syllabus

- 1. Introduction:** Overview, Database System vs. File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure. Data Modelling Using the Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization, Aggregation, Reduction of an ER Diagrams to Tables, Extended ER Model, Relationship of Higher Degree.
- 2. Relational Data Model and Language:** Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Domain Constraints, Relational Algebra, Relational Calculus, Tuple and Domain Calculus. Introduction on SQL: Characteristics of SQL, Advantage of SQL. SQL Data Type and Literals. Types of SQL Commands. SQL Operators and Their Procedure. Tables, Views and Indexes. Queries and Subqueries. Aggregate Functions. Insert, Update and Delete Operations, Joins, Unions, Intersection, Minus, Cursors, Triggers, Procedures in SQL/PL SQL.
- 3. Database Design & Normalization:** Functional dependencies, normal forms, first, second, 8 third normal forms, BCNF, inclusion dependence, lossless join decompositions, normalization using FD, MVD, and JDs, alternative approaches to database design.
- 4. Transaction Processing Concept:** Transaction System, Testing of Serializability, Serializability of Schedules, Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery, Checkpoints, Deadlock Handling. Distributed Database: Distributed Data Storage, Concurrency Control, Directory System. Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Time Stamping Protocols for Concurrency Control, Validation Based Protocol, Multiple Granularity, Multi Version Schemes, Recovery with Concurrent Transaction, Case Study of Oracle.

Course Outcome (CO): At the end of course, the student will be able to

CO1: Apply knowledge of databases for real life applications.

CO2: Apply query processing techniques to automate the real time problems of databases.

CO3: Identify and solve the redundancy problem in database tables using normalization.

CO4: Understand the concepts of transactions, their processing so they will be familiar with broad range of database management issues including data integrity, security and recovery.

References

1. Korth, Silbertz, Sudarshan, "Database Concepts", McGraw Hill.
2. Date C J, "An Introduction to Database Systems", Addison Wesley.

3. Elmasri, Navathe, "Fundamentals of Database Systems", Addison Wesley.
4. O'Neil, Databases, Elsevier Pub.
5. Ramakrishnan "Database Management Systems", McGraw Hill.
6. Leon & Leon, "Database Management Systems", Vikas Publishing House.
7. Bipin C. Desai, "An Introduction to Database Systems", Galgotia Publications.
8. Majumdar & Bhattacharya, "Database Management System", TMH.

Database Management Systems Lab (AIML352)

Credit: 0+1

1. Installing oracle/ MYSQL
2. Creating Entity-Relationship Diagram using case tools.
3. Writing SQL statements Using ORACLE /MYSQL:
 - a. Writing basic SQL SELECT statements.
 - b. Restricting and sorting data.
 - c. Displaying data from multiple tables.
 - d. Aggregating data using group functions.
 - e. Manipulating data.
 - f. Creating and managing tables.
4. Implementing Normalization
5. Creating cursor
6. Creating procedure and functions
7. Creating packages and triggers
8. Design and implementation of payroll processing system
9. Design and implementation of Library Information System
10. Design and implementation of Student Information System
11. Automatic Backup of Files and Recovery of Files
12. Sample of Mini project (Design & Development of Data and Application) for following:
 - a. Inventory Control System.
 - b. Material Requirement Processing.
 - c. Hospital Management System.
 - d. Railway Reservation System.
 - e. Personal Information System.
 - f. Web Based User Identification System.
 - g. Timetable Management System.
 - h. Hotel Management System

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner

Design and Analysis of Algorithms (AIML303)

Credit: 4+0

Course Objective: The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms. At the end of the course students will be able to develop their own versions for a given computational task and to compare and contrast their performance.

Syllabus

1. Introduction: Algorithms, Analyzing Algorithms, Complexity of Algorithms, Growth of Functions, Performance Measurements, Sorting and Order Statistics - Shell Sort, Quick Sort, Merge Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time.
2. Advanced Data Structures: Red-Black Trees, B – Trees, Binomial Heaps, Fibonacci Heaps, Tries, Skip List.
3. Divide and Conquer with Examples Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods with Examples Such as Optimal Reliability Allocation, Knapsack, Minimum Spanning Trees – Prim's and Kruskal's Algorithms, Single Source Shortest Paths - Dijkstra's and Bellman Ford Algorithms.
4. Dynamic Programming with Examples Such as Knapsack. All Pair Shortest Paths – Warshal's and Floyd's Algorithms, Resource Allocation Problem. Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.

Course Outcome (CO): At the end of course, the student will be able to

CO1: Design new algorithms, prove them correct, and analyse their asymptotic and absolute runtime and memory demands.

CO2: Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).

CO3: Understand the mathematical criterion for deciding whether an algorithm is efficient, and know many practically important problems that do not admit any efficient algorithms.

CO4: Apply classical sorting, searching, optimization and graph algorithms.

References

1. Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, "Introduction to Algorithms", Printice Hall of India.
2. E. Horowitz & S Sahni, "Fundamentals of Computer Algorithms",

3. Aho, Hopcraft, Ullman, "The Design and Analysis of Computer Algorithms" Pearson Education, 2008.
4. LEE "Design & Analysis of Algorithms (POD)", McGraw Hill
5. Richard E. Neapolitan "Foundations of Algorithms" Jones & Bartlett Learning
6. Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2005.
7. Michael T Goodrich and Roberto Tamassia, Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Wiley, 2006.
8. Harry R. Lewis and Larry Denenberg, Data Structures and Their Algorithms, Harper Collins, 1997
9. Robert Sedgewick and Kevin Wayne, Algorithms, fourth edition, Addison Wesley, 2011.
10. Harsh Bhasin, "Algorithm Design and Analysis", First Edition, Oxford University Press.
11. Gilles Brassard and Paul Bratley, Algorithmics: Theory and Practice, Prentice Hall, 1995.

Design and Analysis of Algorithm Lab (AIML353)**Credit: 0+1**

1. Program for Recursive Binary & Linear Search.
2. Program for Heap Sort.
3. Program for Merge Sort.
4. Program for Selection Sort.
5. Program for Insertion Sort.
6. Program for Quick Sort.
7. Knapsack Problem using Greedy Solution
8. Perform Travelling Salesman Problem
9. Find Minimum Spanning Tree using Kruskal's Algorithm
10. Implement N Queen Problem using Backtracking
11. Sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of $n > 5000$ and record the time taken to sort. Plot a graph of the time taken versus non graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate using Java how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case.
12. Sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of $n > 5000$, and record the time taken to sort. Plot a graph of the time taken versus non graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case.
13. Implement, the 0/1 Knapsack problem using
 - a. Dynamic Programming method
 - b. Greedy method
14. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.
15. Find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm. Use Union-Find algorithms in your program.
16. Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.
17. Write programs to
 - a. Implement All-Pairs Shortest Paths problem using Floyd's algorithm.
 - b. Implement Travelling Sales Person problem using Dynamic programming.
18. Design and implement to find a subset of a given set $S = \{S_1, S_2, \dots, S_n\}$ of n positive integers whose SUM is equal to a given positive integer d. For example, if $S = \{1, 2, 5, 6, 8\}$ and $d = 9$, there are two solutions $\{1, 2, 6\}$ and $\{1, 8\}$. Display a suitable message, if the given problem instance doesn't have a solution.
19. Design and implement to find all Hamiltonian Cycles in a connected undirected Graph G of n vertices using backtracking principle.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner. It is also suggested that open source tools should be preferred to conduct the lab (C, C++ etc).

Data Mining and Data Warehousing (AIML304)

Credit: 3+0

Course Objective: The course content enables students to: Analyze the difference between On Line Transaction Processing and On-Line analytical processing. To Create Multidimensional schemas suitable for data warehousing. To Understand various data mining functionalities. To Understand in detail about data mining algorithms.

Syllabus

- 1) Data Warehousing and Business Analysis:** Data warehousing Components –Building a Data warehouse –Data Warehouse Architecture – DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools –Metadata – reporting – Query tools and Applications – Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.
- 2) Data Mining:** Data Mining Functionalities – Data Preprocessing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation- Architecture of Typical Data Mining Systems- Classification of Data Mining Systems. **Association Rule Mining:** - Efficient and Scalable Frequent Item set Mining Methods – Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis – Constraint-Based Association Mining.
- 3) Classification and Prediction:** Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Backpropagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods – Prediction – Accuracy and Error Measures – Evaluating the Accuracy of a Classifier or Predictor – Ensemble Methods – Model Section.
- 4) Cluster Analysis, Mining Object, Spatial, Multimedia, Text and Web Data:** Types of Data in Cluster Analysis – A Categorization of Major Clustering Methods – Partitioning Methods – Hierarchical methods – Density-Based Methods – Grid-Based Methods – Model-Based Clustering Methods – Clustering High-Dimensional Data – Constraint-Based Cluster Analysis – Outlier Analysis. Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects – Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the World Wide Web.

Course Outcome (CO): At the end of course, the student will be able to

CO1: To understand the principles of Data Warehousing and Data Mining.

CO2: To be familiar with the Data warehouse architecture and its Implementation.

CO3: To know the Architecture of a Data Mining system.

CO4: To understand the various Data preprocessing Methods.

References

1. Jiawei Han, Micheline Kamber and Jian Pei “Data Mining Concepts and Techniques”, Third Edition, Elsevier, 2011.
2. Alex Berson and Stephen J. Smith “Data Warehousing, Data Mining & OLAP”, Tata McGraw – Hill Edition, Tenth Reprint 2007.
3. K.P. Soman, Shyam Diwakar and V. Ajay “Insight into Data mining Theory and Practice”, Easter Economy Edition, Prentice Hall of India, 2006.

Natural Language Processing (AIML305)

Credit: 3+0

Course Objective: To understand the advanced concepts of Natural Language Processing and to apply the various concepts of NLP in other application areas.

Syllabus

- 1. Introduction:** Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance, WORD LEVEL ANALYSIS : Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.
- 2. Syntactic Analysis:** Context Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing – Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.
- 3. Semantics and Pragmatics:** Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.
- 4. Basic Concepts of Speech Processing:** Speech Fundamentals: Articulatory Phonetics – Production and Classification of Speech Sounds; Acoustic Phonetics – Acoustics of Speech Production; Review of Digital Signal Processing Concepts; Short-Time Fourier Transform, Filter-Bank and LPC Methods. Speech-Analysis: Features, Feature Extraction and Pattern Comparison Techniques: Speech Distortion Measures– Mathematical and Perceptual – Log–Spectral Distance, Cepstral Distances, Weighted Cepstral Distances and Filtering, Likelihood Distortions, Spectral Distortion Using a Warped Frequency Scale, LPC, PLP And MFCC Coefficients, Time Alignment and Normalization – Dynamic Time Warping, Multiple Time – Alignment Paths. SPEECH MODELING: Hidden Markov Models: Markov Processes, HMMs – Evaluation, Optimal State Sequence – Viterbi Search, Baum-Welch Parameter Re-Estimation, Implementation Issues.

Course Outcome (CO): At the end of course, the student will be able

CO1: To learn the fundamentals of natural language processing.

CO2: To understand the use of CFG and PCFG in NLP.

CO3: To understand the role of semantics of sentences and pragmatic.

CO4: To Introduce Speech Production and Related Parameters of Speech.

References

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
2. Steven Bird, Ewan Klein and Edward Loper, —Natural Language Processing with Python, First Edition, O'Reilly Media, 2009.
3. Lawrence Rabiner and Biing-Hwang Juang, “Fundamentals of Speech Recognition”, Pearson Education, 2003. Daniel Jurafsky and James H Martin, “Speech and Language Processing – An Introduction To Natural Language Processing, Computational Linguistics, And Speech Recognition”, Pearson Education, 2002.
4. Frederick Jelinek, “Statistical Methods of Speech Recognition”, MIT Press, 1997.
5. Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher.

Distributed System (AIML306)

Credit: 3+0

Course Objective: This course is an introduction to the design of distributed systems and algorithms that support distributed computing.

Syllabus

- 1. Characterization of Distributed Systems:** Introduction, Examples of distributed Systems, Resource sharing and the Web Challenges. Architectural models, Fundamental Models. Theoretical Foundation for Distributed System: Limitation of Distributed system, absence of global clock, shared memory, Logical clocks, Lamport's & vectors logical clocks. Concepts in Message Passing Systems: causal order, total order, total causal order, Techniques for Message Ordering, Causal ordering of messages, global state, and termination detection.
- 2. Distributed Mutual Exclusion:** Classification of distributed mutual exclusion, requirement of mutual exclusion theorem, Token based and non-token-based algorithms, performance metric for distributed mutual exclusion algorithms. Distributed Deadlock Detection: system model, resource Vs communication deadlocks, deadlock prevention, avoidance, detection & resolution, centralized deadlock detection, distributed deadlock detection, path pushing algorithms, edge chasing algorithms.
- 3. Agreement Protocols:** Introduction, System models, classification of Agreement Problem, Byzantine agreement problem, Consensus problem, Interactive consistency Problem, Solution to Byzantine Agreement problem, Application of Agreement problem, Atomic Commit in Distributed Database system. Distributed Resource Management: Issues in distributed File Systems, Mechanism for building distributed file systems, Design issues in Distributed Shared Memory, Algorithm for Implementation of Distributed Shared Memory.
- 4. Failure Recovery in Distributed Systems & Transactions and Concurrency Control:** Concepts in Backward and Forward recovery, Recovery in Concurrent systems, obtaining consistent Checkpoints, Recovery in Distributed Database Systems. Fault Tolerance: Issues in Fault Tolerance, Commit Protocols, Voting protocols, Dynamic voting protocols. Transactions and Concurrency Control: Transactions, Nested transactions, Locks, Optimistic Concurrency control, Timestamp ordering, Comparison of methods for concurrency control. Distributed Transactions: Flat and nested distributed transactions, Atomic Commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery. Replication: System model and group communication, Fault - tolerant services, highly available services, Transactions with replicated data.

Course Outcome (CO) : At the end of course, the student will be able to

CO1: Provide hardware and software issues in modern distributed systems.

CO2: Get knowledge in distributed architecture, naming, synchronization, consistency and replication, fault tolerance, security, and distributed file systems.

CO3: Analyze the current popular distributed systems such as peer-to-peer (P2P) systems will also be analyzed.

CO4: Know about Shared Memory Techniques and have sufficient knowledge about file access.

References

1. Singhal & Shivaratri, "Advanced Concept in Operating Systems", McGraw Hill
2. Ramakrishna, Gehrke, "Database Management Systems", McGraw Hill
3. Vijay K. Garg Elements of Distributed Computing, Wiley
4. Coulouris, Dollimore, Kindberg, "Distributed System: Concepts and Design", Pearson Education
5. Tenenbaum, Steen, "Distributed Systems", PHI

Neural Network & Deep Learning (AIML307)